AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M238 : Data Visualization

Final Examination
Spring Semester 2024/2025

Date: 12/June/2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **Three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1

TRUE or FALSE (10 marks)

MARK the following statements are TRUE or FALSE

1. Datasets provide essential information that can be visualized effectively.
2. To visualize geospatial data in context, maps are typically created with multiple layers displaying different types of information.
3. In geographic coordinates, longitude, latitude, and altitude are specified independently of any reference system.
4. Flattening the Earth's spherical surface into a flat map is achieved through a process called projection.
5. Color is an ineffective tool in enhancing data visualizations.
6. Qualitative color scales work best when they are applied to only one category of data.
7. For large and complex datasets, creating a single figure panel is always the best approach.
8. Compound figures consist of separate figure panels arranged in an arbitrary (non-grid) layout.
9. Color can be used to represent numerical data values, such as temperature or speed, in visualizations.
10. A story in data visualization is defined as a set of observations, facts, or events arranged in a specific order to evoke an emotional reaction.

PART 2

ALL QUESTIONS ARE REQUIRED (30 marks)

Question 1: The equator divides the Earth into two hemispheres. To uniquely specify a location on Earth, list the three pieces of information required and explain the role of each ? (3 Marks)

Question 2: Define a choropleth map and provide one example of how it might be used. (4 Marks)

Question 3: What is a cartogram? Explain how it differs from a conventional geographic map. (4 Marks)

Question 4: Explain the difference between vector graphics and bitmap (raster) graphics, including one key advantage of each. (5 Marks)

Question 5: How many distinct categories are recommended for qualitative color scales in data visualizations, and why? (4 Marks)

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Question 6: Explain how storytelling with data visualizations can enhance audience engagement. (2 Marks)

Question 7: Compare lossy and lossless compression in image formats. Which is better for data visualizations, and why? (5 Marks)

Question 8: List the key steps to make a visualization memorable. (3 Marks)

PART 3

ALL QUESTIONS ARE REQUIRED **(10 marks)**

Write Python code to generate a scatter plot visualizing the relationship between head length and body mass. Ensure your code:

- **Uses matplotlib to create a figure of size (8, 6).**
- **Plots the data with blue circular markers.**
- **Includes a title (Head Length vs. Body Mass), labeled axes, and grid lines.**

, figure1. (10 marks)

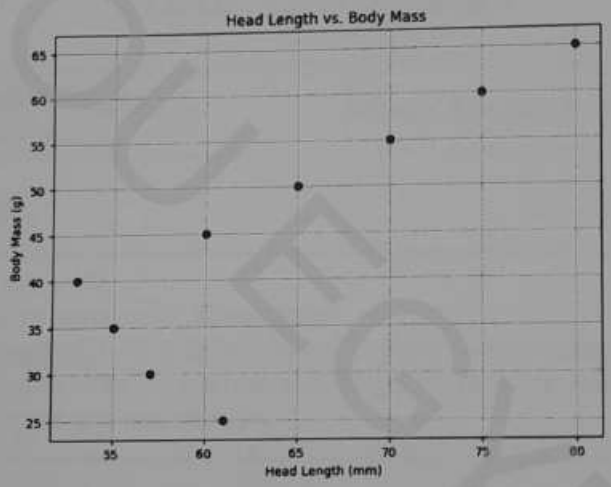


Figure 1

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End of questions

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